

Simulation Atlas

Extended sensitivity analyses for The Link Function Problem in Psychological Interaction Testing

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Table of contents

1	Purpose and scope	1
1.1	How to read the figures	2
2	Atlas design	2
3	Simulation 1: Forced-choice accuracy and the non-zero chance floor	3
3.1	Manuscript anchors	3
3.2	One-dimensional sensitivity slices	6
3.3	Main-effect surface	7
4	Simulation 2: Within-family link choice, logit versus probit	10
4.1	Manuscript anchors	10
4.2	One-dimensional sensitivity slices	11
4.3	Main-effect surface	15
5	Simulation 3: Sum scores	15
5.1	Manuscript anchors	15
5.2	One-dimensional sensitivity slices	16
5.3	Main-effect surface	19
6	Diagnostic atlas	21
6.1	Detection across sensitivity slices	22
6.2	Strength of the AIC comparison	22
7	What the atlas adds, and what it does not	25

8 Numerical appendix	26
8.1 Forced-choice accuracy	26
8.2 Logit versus probit	35
8.3 Sum scores	42

1 Purpose and scope

This document reports the extended simulation atlas for *The Link Function Problem in Psychological Interaction Testing*. It follows the terminology and simulation numbering of the manuscript version dated 2026-08-11. The manuscript keeps a small number of transparent worked scenarios. The atlas asks how far those results extend when selected design and data-generating parameters are varied around the manuscript anchors. It is designed to be read independently of the interactive application: the relevant design settings, coefficients, fitted models, and outcome summaries are reported here, while references to the manuscript identify the corresponding conceptual examples.

The atlas is a structured sensitivity analysis, not an exhaustive map of psychological settings. It uses prespecified one-dimensional slices and a small main-effect surface instead of a full Cartesian product. In every scenario, the interaction term is **exactly zero on the generating scale**. Therefore, a rejection under the matched generating model estimates the nominal generating-scale rejection rate, while a rejection under another scale is described as a **pseudo-interaction detection rate**. As in the manuscript, *pseudo-interaction* is a simulation term. It is available here because the generating scale is known.

The current compact summaries come from atlas run type full, atlas version 0.1.0. The atlas contains 93 unique generating scenarios: 24 for forced-choice accuracy, 46 for the logit-probit comparison, and 23 for sum scores. The forced-choice and sum-score scenarios use 500 Monte Carlo replications per scenario; the more expensive within-family GLMM scenarios use 300.

1.1 How to read the figures

All rejection-rate figures use the same interpretation. The horizontal dashed line at .05 marks nominal alpha. Rates close to .05 under the generating scale provide a calibration check. Elevated rates under an alternative fitted scale show how often the scale difference becomes a statistically significant product term at the stated sample size and design. Error bars are shown for the manuscript anchors. The sensitivity plots omit error bars to keep the larger set of scenarios readable; Wilson 95% intervals are available in the numerical tables and in `data/atlas-summary.csv`.

The deterministic pseudo-interaction reported in the tables is computed from the expected values before sampling. It describes the interaction contrast implied by evaluating the gener-

ating pattern on the fitted scale. The Monte Carlo rejection rate adds sampling variability, standard errors, and sample size to that deterministic discrepancy.

2 Atlas design

Table 1: Structure of the simulation atlas. Each family contains the manuscript anchor scenarios plus prespecified sensitivity slices.

Manuscript case	Generating scenarios	Monte Carlo replications	Sensitivity slices
Simulation 1: Forced-choice accuracy	24	500	Main effects; sample size; scale location; trials; chance level
Simulation 2: Logit vs probit	46	300	Main effects; sample size; trials; latent location; ICC
Simulation 3: Sum scores	23	500	Main effects; sample size; scale location; item count; latent dispersion

The one-dimensional slices change one named parameter while holding the remaining settings at the central manuscript anchor. The main-effect surfaces vary the two additive main-effect coefficients jointly. The actual coefficients are reported directly on the figure axes and in the table below. Duplicate anchor combinations are stored once and can belong to several slices.

Table 2: Generating-scale coefficients used in the main-effect surfaces. Coefficients are reported on the scale that generated each scenario.

Manuscript case	Generating link/scale	First main effect	First coefficient values	Second main effect	Second coefficient values
Simulation 1: Forced-choice accuracy	Chance-corrected logit	Age coefficient	0.30, 0.60, 0.90	Group coefficient	-1.35, -0.90, -0.45
Simulation 2: Logit vs probit	Logit	Group coefficient	-2.475, -1.650, -0.825	Condition coefficient	-2.475, -1.650, -0.825
Simulation 2: Logit vs probit	Probit	Group coefficient	-1.50, -1.00, -0.50	Condition coefficient	-1.50, -1.00, -0.50

Table 2: Generating-scale coefficients used in the main-effect surfaces. Coefficients are reported on the scale that generated each scenario.

Manuscript case	Generating link/scale	First main effect	First coefficient values	Second main effect	Second coefficient values
Simulation 3: Sum scores	Latent generating scale	Predictor coefficient	0.425, 0.850, 1.275	Group coefficient	-1.35, -0.90, -0.45

For Simulation 2, logit and probit coefficients are on different generating-link scales. As in the manuscript simulation, the logit coefficients are scaled by 1.65 relative to the probit reference coefficients so that the two generating scenarios have similar cell probabilities. The figures therefore report the actual coefficient values for each generating link instead of a common relative index.

3 Simulation 1: Forced-choice accuracy and the non-zero chance floor

The manuscript generates accuracy from a chance-corrected binomial logit model with a .50 floor and no age-by-group product term on that scale. It compares the matched chance-corrected model with a standard binomial logit, a standard binomial probit, and a Gaussian identity model. The atlas keeps this comparison and extends the paper scenarios across sample size, overall performance, number of trials, chance level, and the joint values of the age and group coefficients. In the main-effect surface, the generating-scale age coefficients are 0.30, 0.60, 0.90 and the group coefficients are -1.35, -0.90, -0.45.

3.1 Manuscript anchors

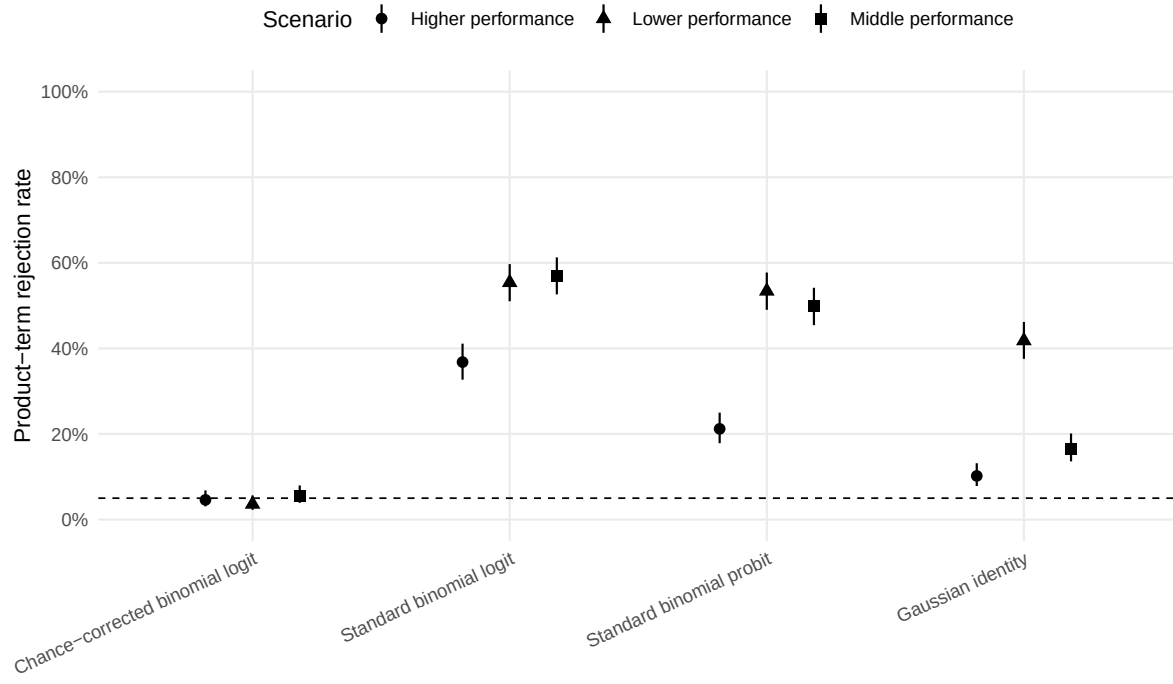


Figure 1: Forced-choice manuscript anchors reproduced within the atlas. The product term is zero on the chance-corrected logit generating scale. The dashed line marks $\alpha = .05$.

Table 3: Forced-choice manuscript anchors. The deterministic pseudo-interaction is evaluated on each fitted scale before sampling.

Scenario	Generating link/scale	Fitted model/scale	Rate type	Rejection rate	95% CI	Deterministic pseudo-interaction	Fit success
Lower performance	chance-corrected logit	Chance-corrected binomial logit	Nominal generating-scale rejection rate	3.6%	[2.3%, 5.6%]	0.000	100.0%
Lower performance	chance-corrected logit	Gaussian identity	Pseudo-interaction detection rate	41.8%	[37.6%, 46.2%]	-0.077	100.0%
Lower performance	chance-corrected logit	Standard binomial logit	Pseudo-interaction detection rate	55.4%	[51.0%, 59.7%]	-0.453	100.0%
Lower performance	chance-corrected logit	Standard binomial probit	Pseudo-interaction detection rate	53.4%	[49.0%, 57.7%]	-0.262	100.0%
Middle performance	chance-corrected logit	Chance-corrected binomial logit	Nominal generating-scale rejection rate	5.6%	[3.9%, 8.0%]	0.000	100.0%
Middle performance	chance-corrected logit	Gaussian identity	Pseudo-interaction detection rate	16.6%	[13.6%, 20.1%]	-0.036	100.0%
Middle performance	chance-corrected logit	Standard binomial logit	Pseudo-interaction detection rate	57.0%	[52.6%, 61.3%]	-0.473	100.0%
Middle performance	chance-corrected logit	Standard binomial probit	Pseudo-interaction detection rate	49.8%	[45.4%, 54.2%]	-0.243	100.0%
Higher performance	chance-corrected logit	Chance-corrected binomial logit	Nominal generating-scale rejection rate	4.6%	[3.1%, 6.8%]	0.000	100.0%
Higher performance	chance-corrected logit	Gaussian identity	Pseudo-interaction detection rate	10.2%	[7.8%, 13.2%]	0.028	100.0%

Table 3: Forced-choice manuscript anchors. The deterministic pseudo-interaction is evaluated on each fitted scale before sampling.

Scenario	Generating link/scale	Fitted model/scale	Rate type	Rejection rate	95% CI	Deterministic pseudo-interaction	Fit success
Higher performance	chance-corrected logit	Standard binomial logit	Pseudo-interaction detection rate	36.8%	[32.7%, 41.1%]	-0.396	100.0%
Higher performance	chance-corrected logit	Standard binomial probit	Pseudo-interaction detection rate	21.2%	[17.8%, 25.0%]	-0.153	100.0%

Across all forced-choice atlas scenarios, the matched chance-corrected model remains close to nominal alpha, with rejection rates from 2.8% to 7.0%. The standard binomial logit ranges from 9.2% to 98.4%, and the standard binomial probit from 7.4% to 97.8%. These ranges are sensitivity results, not prevalence estimates. They show that the inferential consequence of placing the lower asymptote at 0 instead of the task-implied chance level can be small in some regions and very large in others.

3.2 One-dimensional sensitivity slices

Table 4: Range of rejection rates across each forced-choice sensitivity slice.

Slice	Generating link/scale	Fitted model/scale	Minimum rejection rate	Maximum rejection rate
Chance level	chance-corrected logit	Chance-corrected binomial logit	4.8%	6.0%
Sample size	chance-corrected logit	Chance-corrected binomial logit	3.6%	5.6%
Scale location	chance-corrected logit	Chance-corrected binomial logit	3.6%	5.6%
Trials per participant	chance-corrected logit	Chance-corrected binomial logit	5.0%	7.0%
Chance level	chance-corrected logit	Standard binomial logit	48.8%	57.0%
Sample size	chance-corrected logit	Standard binomial logit	16.0%	98.4%

Table 4: Range of rejection rates across each forced-choice sensitivity slice.

Slice	Generating link/scale	Fitted model/scale	Minimum rejection rate	Maximum rejection rate
Scale location	chance-corrected logit	Standard binomial logit	15.0%	57.0%
Trials per participant	chance-corrected logit	Standard binomial logit	22.0%	89.4%
Chance level	chance-corrected logit	Standard binomial probit	45.6%	49.8%
Sample size	chance-corrected logit	Standard binomial probit	13.8%	96.2%
Scale location	chance-corrected logit	Standard binomial probit	7.4%	53.4%
Trials per participant	chance-corrected logit	Standard binomial probit	19.2%	84.0%
Chance level	chance-corrected logit	Gaussian identity	16.6%	25.4%
Sample size	chance-corrected logit	Gaussian identity	5.6%	42.4%
Scale location	chance-corrected logit	Gaussian identity	10.2%	66.2%
Trials per participant	chance-corrected logit	Gaussian identity	8.2%	34.2%

The slices separate distinct reasons why a deterministic scale discrepancy may or may not become a statistical interaction. Scale location and chance level change the curvature relevant to the fitted link. Sample size and number of trials change how precisely that discrepancy is estimated. The figure therefore distinguishes the existence of a pseudo-interaction on the fitted scale from the probability that a product-term test detects it.

3.3 Main-effect surface

The surface directly targets the mechanism emphasized in the manuscript. Two additive main effects can jointly move observations through regions where equal differences on the generating scale map to unequal differences on the fitted scale. Larger main effects can therefore increase the deterministic pseudo-interaction even though the generating age-by-group term remains zero.

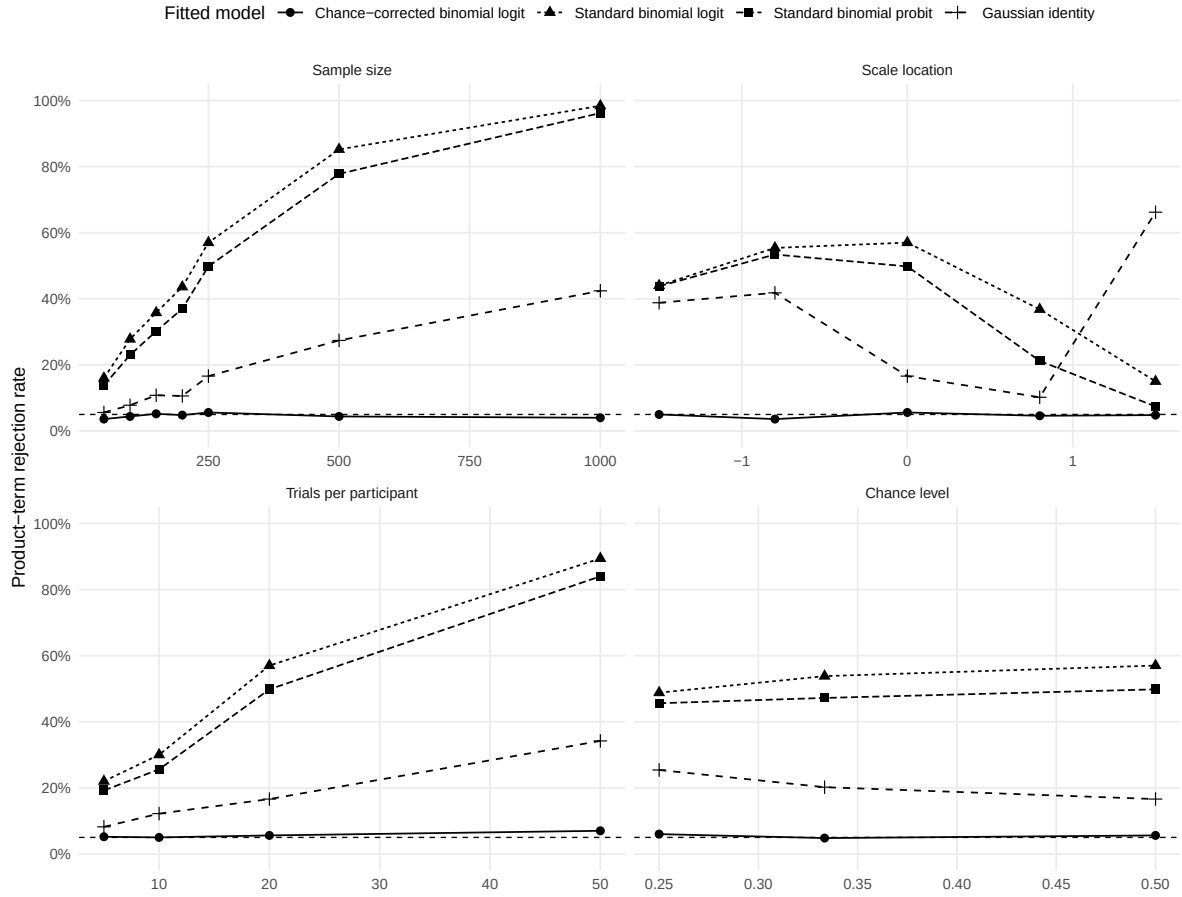


Figure 2: Forced-choice sensitivity slices. One parameter varies at a time around the central manuscript anchor. The dashed line marks $\alpha = .05$.

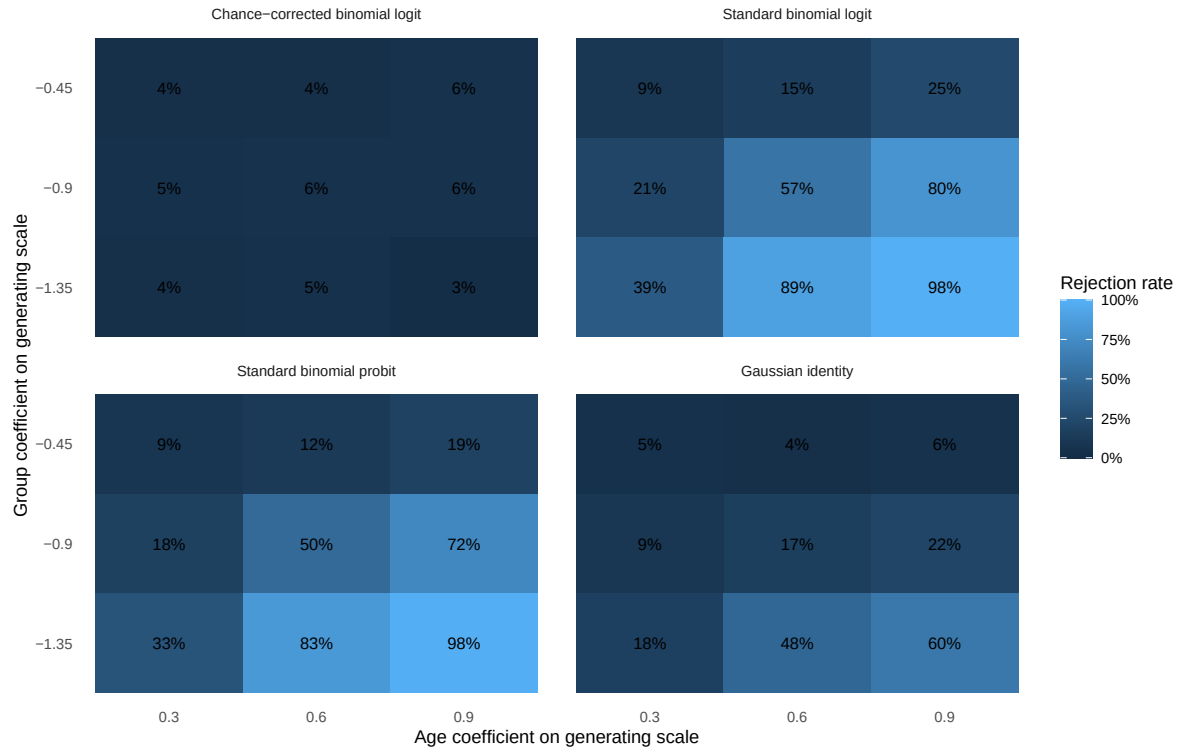


Figure 3: Forced-choice main-effect surface. Axes report the age and group coefficients on the chance-corrected logit generating scale. Cell labels are product-term rejection rates.

4 Simulation 2: Within-family link choice, logit versus probit

The manuscript crosses logit and probit as generating and fitted links in repeated-trial random-intercept GLMMs. Both links use the binomial family and the same 0-to-1 bounds. The difference is the scale on which additivity is imposed. The atlas extends this comparison across sample size, trials per participant-condition cell, latent location, latent ICC, and the joint values of the group and condition coefficients. In the main-effect surface, the probit generating coefficients for both effects are -1.50, -1.00, -0.50, whereas the corresponding logit coefficients are -2.475, -1.650, -0.825.

4.1 Manuscript anchors

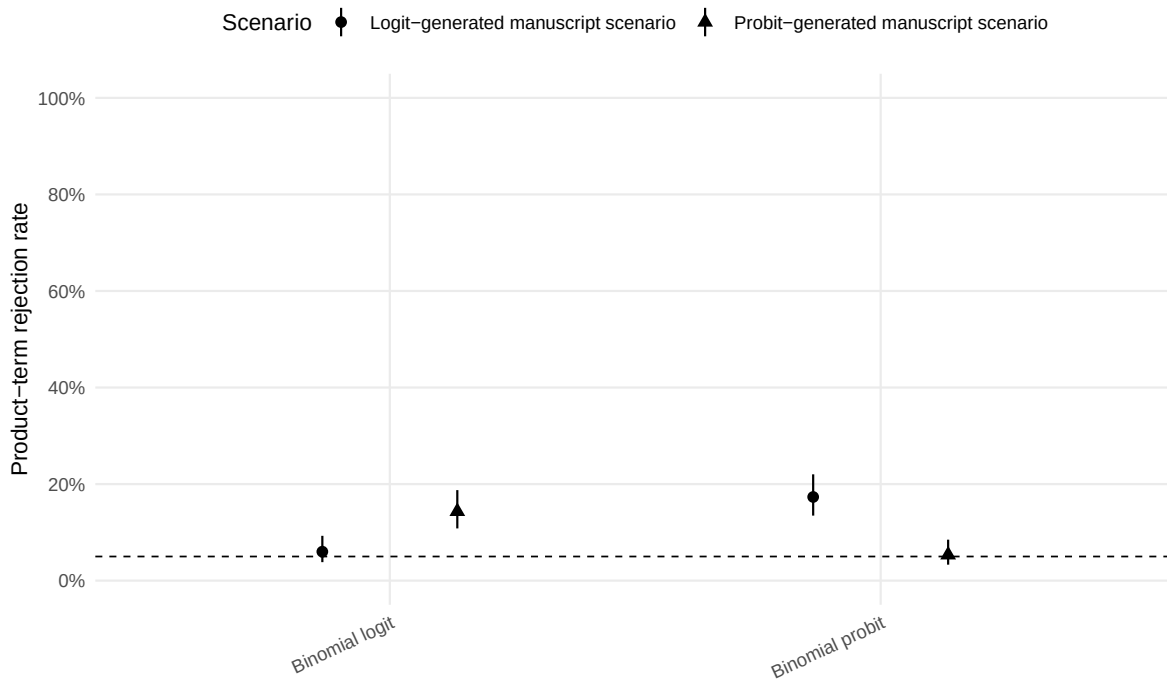


Figure 4: Logit-probit manuscript anchors reproduced within the atlas. The product term is zero on the generating link scale. The dashed line marks $\alpha = .05$.

Table 5: Within-family manuscript anchors for logit and probit generating links.

Scenario	Generating link/scale	Fitted model/scale	Rate type	Rejection rate	95% CI	Deterministic pseudo-interaction	Fit success
Logit-generated manuscript scenario	logit	Binomial logit	Nominal generating-scale rejection rate	6.0%	[3.8%, 9.3%]	0.000	100.0%
Logit-generated manuscript scenario	logit	Binomial probit	Pseudo-interaction detection rate	17.3%	[13.5%, 22.0%]	-0.112	100.0%
Probit-generated manuscript scenario	probit	Binomial logit	Pseudo-interaction detection rate	14.3%	[10.8%, 18.8%]	0.216	100.0%
Probit-generated manuscript scenario	probit	Binomial probit	Nominal generating-scale rejection rate	5.3%	[3.3%, 8.5%]	0.000	100.0%

Across the atlas, matched logit/probit fits remain near the nominal rejection rate. The wrong-link rows range from 5.3% to 82.3%. The range is wider than the manuscript anchor alone suggests because the atlas deliberately varies information and the region of the latent scale. The result remains conditional on these declared slices and should not be read as a general frequency for logit-probit disagreement.

4.2 One-dimensional sensitivity slices

Table 6: Range of rejection rates across each within-family sensitivity slice.

Slice	Generating link/scale	Fitted model/scale	Minimum rejection rate	Maximum rejection rate
Latent ICC	logit	Binomial logit	4.7%	7.3%
Latent location	logit	Binomial logit	5.0%	6.7%
Sample size	logit	Binomial logit	2.7%	7.7%

Table 6: Range of rejection rates across each within-family sensitivity slice.

Slice	Generating link/scale	Fitted model/scale	Minimum rejection rate	Maximum rejection rate
Trials per cell	logit	Binomial logit	4.0%	6.0%
Latent ICC	probit	Binomial logit	8.3%	82.3%
Latent location	probit	Binomial logit	7.7%	44.3%
Sample size	probit	Binomial logit	7.7%	43.0%
Trials per cell	probit	Binomial logit	9.7%	31.7%
Latent ICC	logit	Binomial probit	8.7%	79.0%
Latent location	logit	Binomial probit	9.0%	46.0%
Sample size	logit	Binomial probit	7.3%	37.0%
Trials per cell	logit	Binomial probit	7.3%	28.3%
Latent ICC	probit	Binomial probit	3.0%	6.3%
Latent location	probit	Binomial probit	3.7%	6.0%
Sample size	probit	Binomial probit	4.0%	5.7%
Trials per cell	probit	Binomial probit	5.3%	6.0%

Sample size and trial count mainly change how detectable the small systematic discrepancy between the links becomes. Latent location and the main-effect magnitudes change the discrepancy itself by moving cell probabilities into regions where logit and probit curvature differs more. ICC changes the information structure of the repeated observations while leaving the fixed generating interaction at zero.

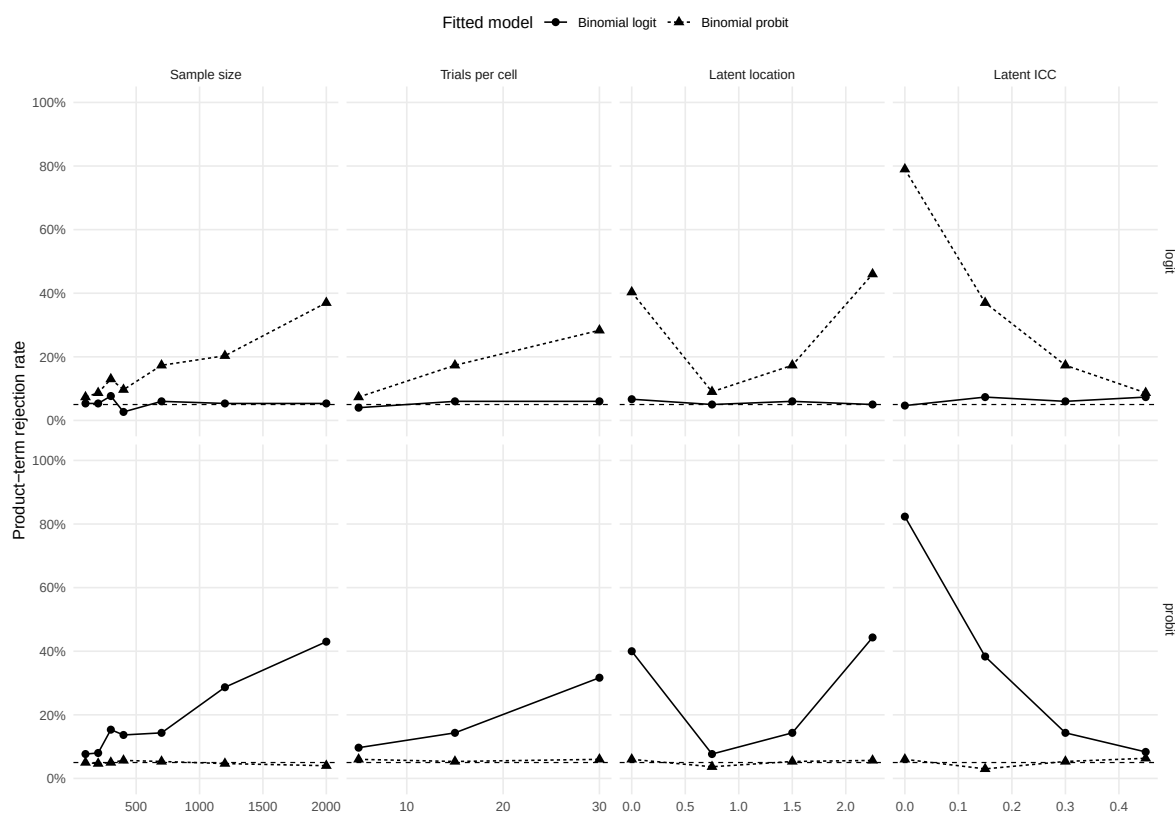


Figure 5: Logit-probit sensitivity slices. Rows identify the generating link. Within each panel the fitted logit and probit models are compared as one parameter varies around the manuscript anchor.

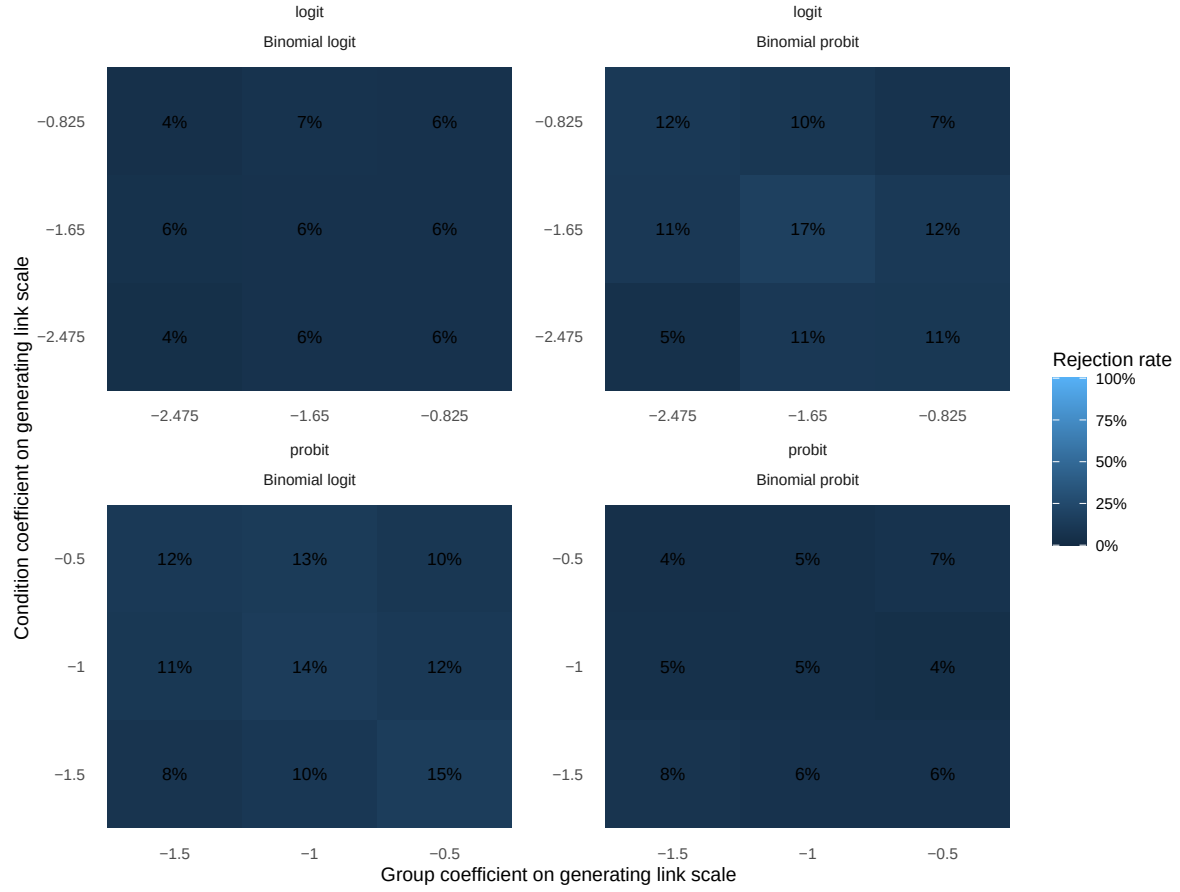


Figure 6: Within-family main-effect surfaces. Axes report the group and condition coefficients on each generating-link scale. Panels identify the generating and fitted links. Cell labels are product-term rejection rates.

4.3 Main-effect surface

The matched-link panels provide the calibration reference. The off-diagonal panels show when the small difference between two standard binomial links becomes consequential for an interaction claim. This is the narrow point of the within-family example: logit and probit can remain very similar for prediction while defining sufficiently different additivity baselines to change a product-term conclusion under high information or strong curvature.

5 Simulation 3: Sum scores

The manuscript treats sum scores as a measurement case. Data are generated on a latent continuous scale with no x-by-group interaction, converted into ordinal item responses, and summed. The atlas compares the latent generating scale, the observed sum-score identity model, and the Gaussian-probit bounded-score sensitivity model. It extends the manuscript scenarios across sample size, item-threshold location, number of items, latent dispersion, and the joint values of the predictor and group coefficients. In the main-effect surface, the latent-scale predictor coefficients are 0.425, 0.850, 1.275 and the group coefficients are -1.35, -0.90, -0.45.

5.1 Manuscript anchors

Table 7: Sum-score manuscript anchors. The latent generating scale is available only in simulation.

Scenario	Generating link/scale	Fitted model/scale	Rate type	Rejection rate	95% CI	Deterministic pseudo-interaction	Fit success
Lower range	graded logistic latent scale	Gaussian-probit bounded score	Pseudo-interaction detection rate	6.6%	[4.7%, 9.1%]	-0.056	100.0%
Lower range	graded logistic latent scale	Latent generating scale	Nominal generating-scale rejection rate	3.6%	[2.3%, 5.6%]	0.000	100.0%
Lower range	graded logistic latent scale	Observed sum-score identity	Pseudo-interaction detection rate	72.6%	[68.5%, 76.3%]	-2.034	100.0%

Table 7: Sum-score manuscript anchors. The latent generating scale is available only in simulation.

Scenario	Generating link/scale	Fitted model/scale	Rate type	Rejection rate	95% CI	Deterministic pseudo- interaction	Fit suc- cess
Middle range	graded logistic latent scale	Gaussian-probit bounded score	Pseudo-interaction detection rate	3.8%	[2.4%, 5.9%]	-0.010	100.0%
Middle range	graded logistic latent scale	Latent generating scale	Nominal generating-scale rejection rate	5.8%	[4.1%, 8.2%]	0.000	100.0%
Middle range	graded logistic latent scale	Observed sum-score identity	Pseudo-interaction detection rate	7.4%	[5.4%, 10.0%]	-0.696	100.0%
Upper range	graded logistic latent scale	Gaussian-probit bounded score	Pseudo-interaction detection rate	5.6%	[3.9%, 8.0%]	0.027	100.0%
Upper range	graded logistic latent scale	Latent generating scale	Nominal generating-scale rejection rate	4.4%	[2.9%, 6.6%]	0.000	100.0%
Upper range	graded logistic latent scale	Observed sum-score identity	Pseudo-interaction detection rate	37.6%	[33.5%, 41.9%]	1.570	100.0%

Across the atlas, the latent generating scale rejects from 3.0% to 7.0%. The observed sum-score identity model ranges from 2.8% to 83.2%, while the Gaussian-probit bounded-score sensitivity model ranges from 2.6% to 12.8%. The latter remains a model of the manifest total. It does not recover the latent construct and should be read as a sensitivity analysis for the scale on which additivity is imposed.

5.2 One-dimensional sensitivity slices

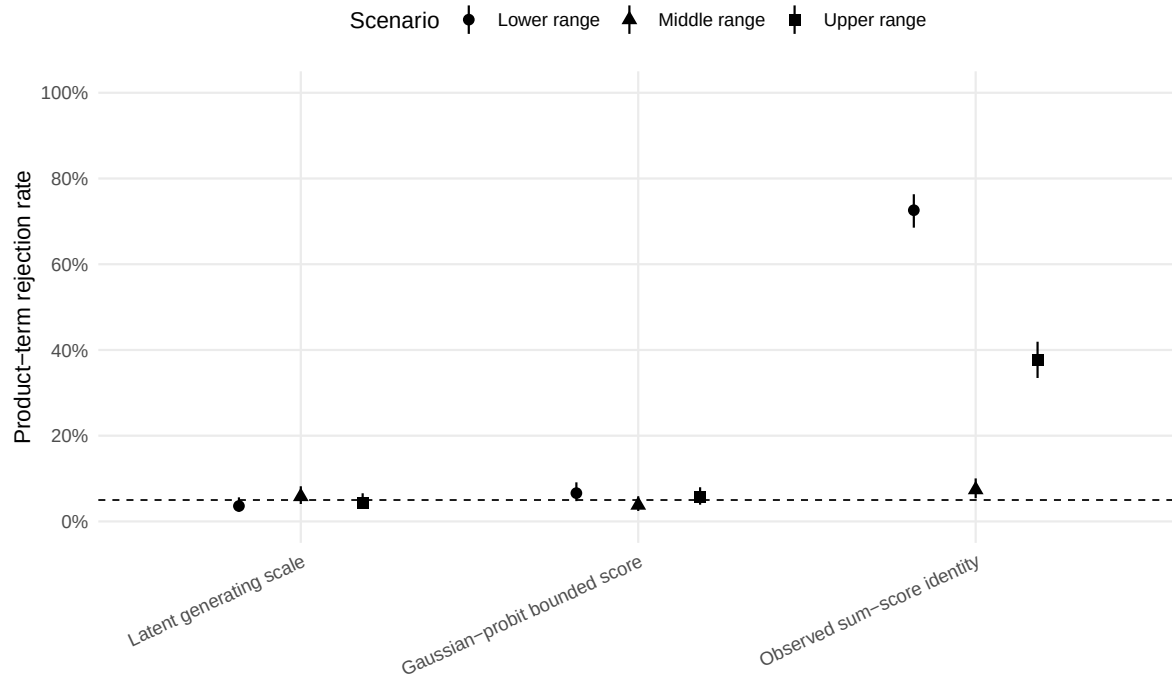


Figure 7: Sum-score manuscript anchors reproduced within the atlas. The product term is zero on the latent generating scale. The dashed line marks $\alpha = .05$.

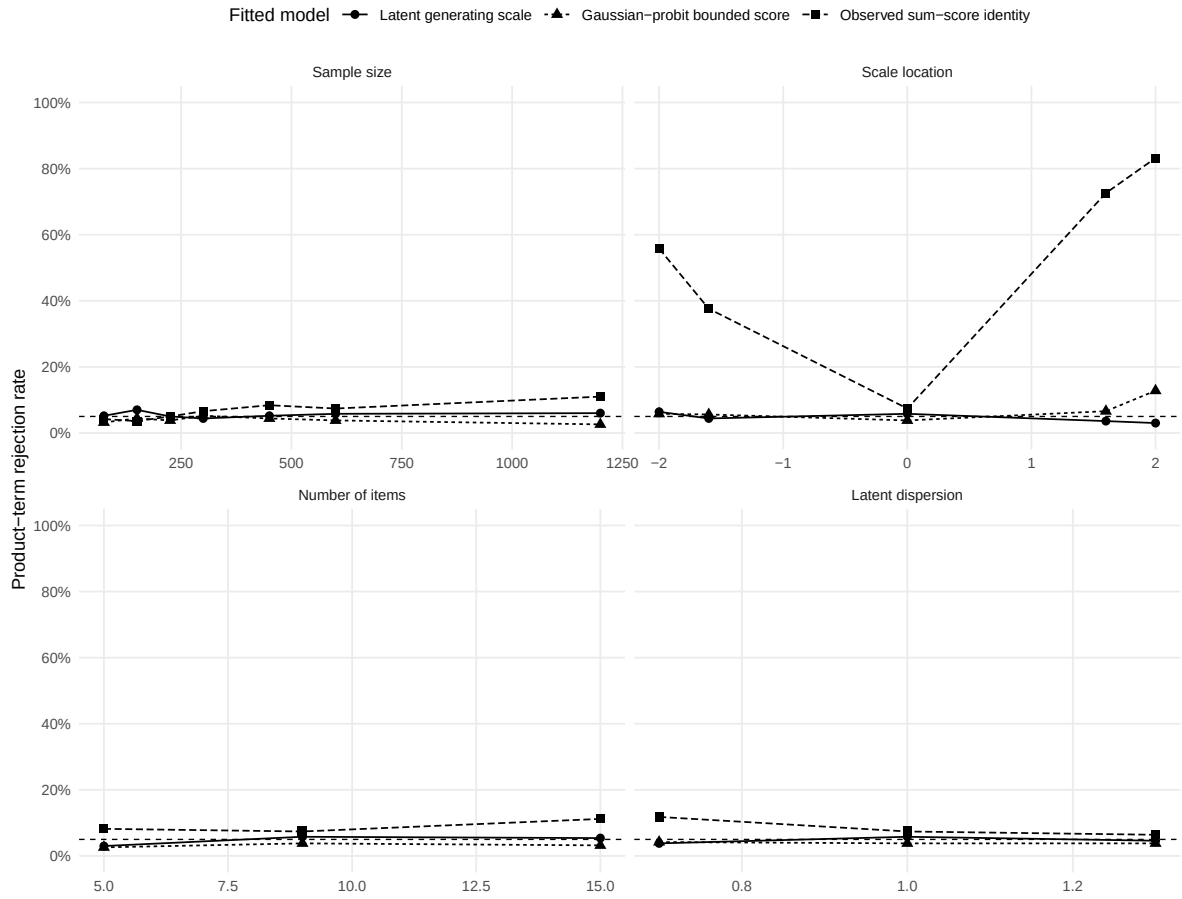


Figure 8: Sum-score sensitivity slices. One parameter varies at a time around the central manuscript anchor. The dashed line marks $\alpha = .05$.

Table 8: Range of rejection rates across each sum-score sensitivity slice.

Slice	Generating link/scale	Fitted model/scale	Minimum rejection rate	Maximum rejection rate
Latent dispersion	graded logistic latent scale	Latent generating scale	3.8%	5.8%
Number of items	graded logistic latent scale	Latent generating scale	3.0%	5.8%
Sample size	graded logistic latent scale	Latent generating scale	4.4%	7.0%
Scale location	graded logistic latent scale	Latent generating scale	3.0%	6.4%
Latent dispersion	graded logistic latent scale	Gaussian-probit bounded score	3.8%	4.2%
Number of items	graded logistic latent scale	Gaussian-probit bounded score	2.6%	3.8%
Sample size	graded logistic latent scale	Gaussian-probit bounded score	2.6%	5.2%
Scale location	graded logistic latent scale	Gaussian-probit bounded score	3.8%	12.8%
Latent dispersion	graded logistic latent scale	Observed sum-score identity	6.4%	11.8%
Number of items	graded logistic latent scale	Observed sum-score identity	7.4%	11.2%
Sample size	graded logistic latent scale	Observed sum-score identity	3.6%	11.0%
Scale location	graded logistic latent scale	Observed sum-score identity	7.4%	83.2%

The scale-location slice changes how strongly the bounded sum score compresses the latent dimension. Item count and latent dispersion change the mapping between the latent variable and the manifest total. Sample size changes the probability that any resulting fitted-scale interaction is detected. These are distinct components of the sum-score problem and should not be reduced to a generic floor-or-ceiling warning.

5.3 Main-effect surface

The surface shows how the mapping from latent values to a bounded total can turn two additive latent effects into a non-additive pattern on the manifest score. The latent benchmark remains

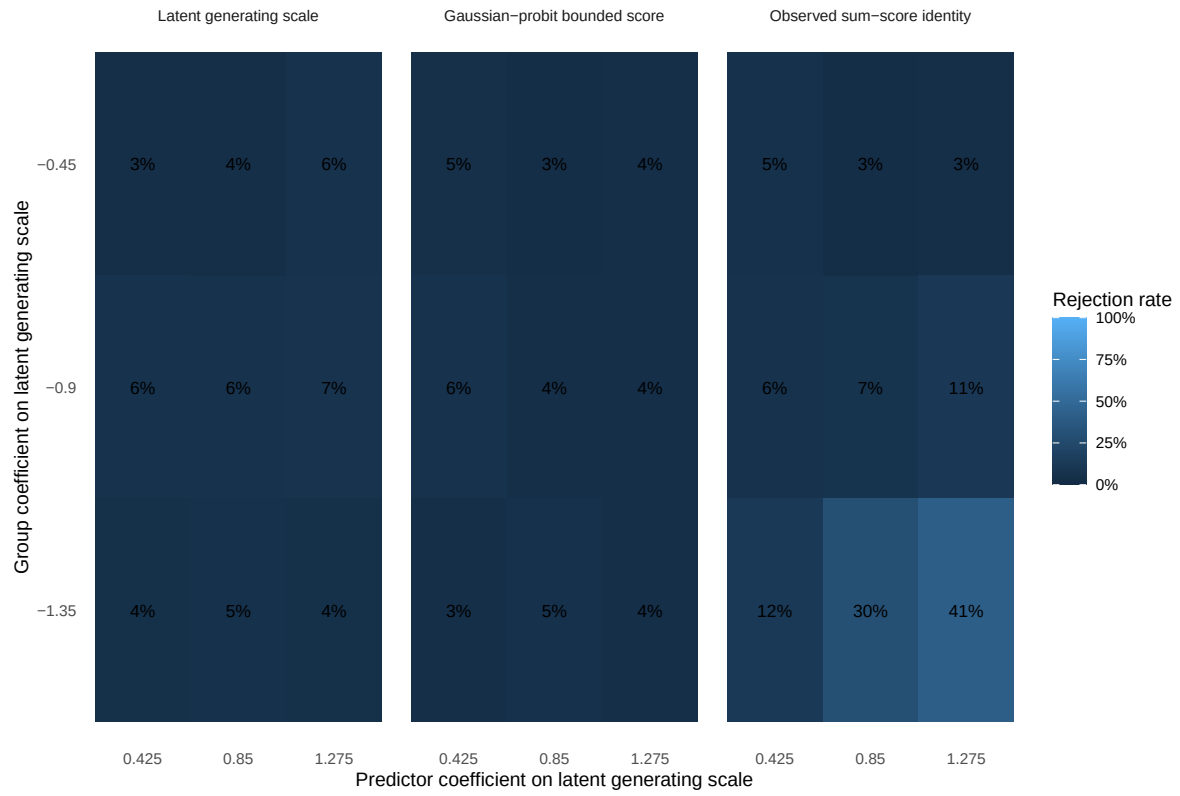


Figure 9: Sum-score main-effect surface. Axes report the predictor and group coefficients on the latent generating scale. Cell labels are product-term rejection rates.

the calibration reference because its product term is zero by construction.

6 Diagnostic atlas

The diagnostic atlas extends two of the manuscript’s five diagnostic scenarios: the forced-choice chance-floor case and the probit-generated, logit-fitted repeated-trial case. It is narrower than the main diagnostic simulation in the paper. Its purpose is to ask whether the relation between pseudo-interaction risk and diagnostic detection changes as sample size and scale location vary.

Diagnostics are applied to the **misspecified interaction model**, matching the manuscript’s post-estimation question: once the fitted model has produced the product-term finding, does that same model show enough evidence of misspecification to warn the analyst? AIC compares prespecified candidate mean models with the same structural interaction formula. Positive delta AIC, defined as AIC of the misspecified model minus AIC of the generating model, favors the generating link.

Table 9: Diagnostic coverage in the current atlas summary. Applicable and computed are separate: a check can be structurally applicable but deferred in the current run.

Case	Diagnostic	Applicable	Computed
Forced-choice chance-floor case	AIC favors generating link	Yes	Yes
Probit-generated, logit-fitted case	AIC favors generating link	Yes	Yes
Forced-choice chance-floor case	DHARMa dispersion	Yes	No
Probit-generated, logit-fitted case	DHARMa dispersion	Yes	No
Forced-choice chance-floor case	DHARMa residual distribution across design cells	No	No
Probit-generated, logit-fitted case	DHARMa residual distribution across design cells	Yes	No
Forced-choice chance-floor case	DHARMa residual quantiles over fitted values	Yes	No
Probit-generated, logit-fitted case	DHARMa residual quantiles over fitted values	No	No
Forced-choice chance-floor case	DHARMa residual quantiles over focal predictor	Yes	No

Table 9: Diagnostic coverage in the current atlas summary. Applicable and computed are separate: a check can be structurally applicable but deferred in the current run.

Case	Diagnostic	Applicable	Computed
Probit-generated, logit-fitted case	DHARMa residual quantiles over focal predictor	No	No
Forced-choice chance-floor case	DHARMa uniformity	Yes	No
Probit-generated, logit-fitted case	DHARMa uniformity	Yes	No
Forced-choice chance-floor case	Pregibon-style added-term link check	Yes	Yes
Probit-generated, logit-fitted case	Pregibon-style added-term link check	Yes	Yes

The current diagnostic summary does not yet contain computed DHARMa results. Those checks are retained as applicable but not computed, so they are not treated as negative findings. If the diagnostic atlas is re-run with DHARMa enabled and the summary files are replaced, the figure below will include the additional computed checks automatically.

6.1 Detection across sensitivity slices

The rates in this figure answer different questions. Pseudo-interaction detection is the probability that the misspecified fitted scale yields a significant product term. Pregibon-style detection is the probability that the added-term link check flags a departure. AIC reports how often the generating link has the lower AIC within the specified candidate pair. The curves therefore need not be complements of one another. Diagnostic misalignment occurs when the interaction conclusion changes more readily than the diagnostic workflow signals a problem.

6.2 Strength of the AIC comparison

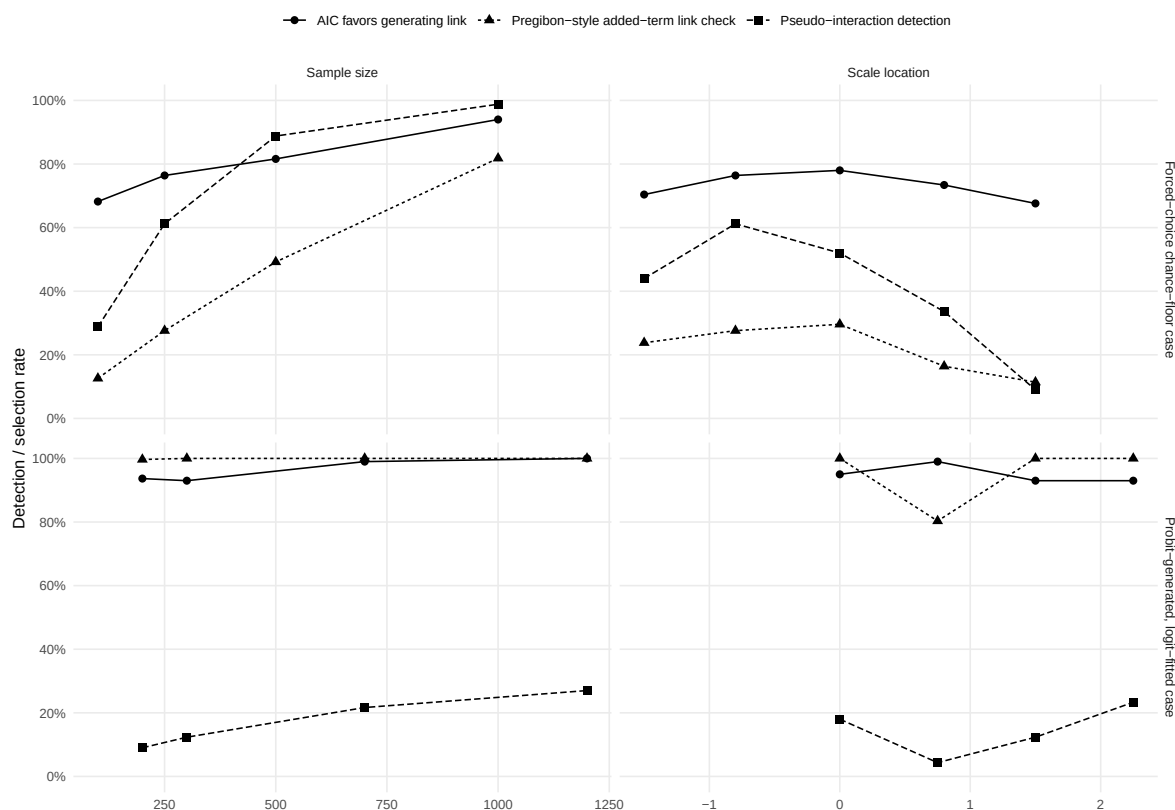


Figure 10: Diagnostic atlas. Pseudo-interaction detection is shown together with each diagnostic or candidate-model comparison computed in the current run. Sample-size and scale-location slices are shown separately.

Table 10: AIC comparison across the diagnostic atlas. Positive delta AIC favors the generating link. The within-two column reports the proportion of finite comparisons with absolute delta AIC below 2.

Case	Slice	Value	Pseudo- interaction detection	AIC favors generating link	Median delta AIC	Delta AIC IQR	Absolute delta AIC < 2
Forced-choice chance-floor case	Sample size	100.00	28.8%	68.2%	0.88	[-0.32, 2.22]	62.8%
Forced-choice chance-floor case	Sample size	250.00	61.2%	76.4%	2.25	[0.19, 4.28]	37.4%
Forced-choice chance-floor case	Sample size	500.00	88.8%	81.6%	4.39	[1.32, 7.51]	21.6%
Forced-choice chance-floor case	Sample size	1000.00	98.8%	94.0%	9.30	[5.23, 13.27]	7.6%
Probit- generated, logit-fitted case	Sample size	200.00	9.0%	93.7%	4.21	[2.15, 6.10]	22.0%
Probit- generated, logit-fitted case	Sample size	300.00	12.3%	93.0%	6.13	[3.81, 8.99]	13.3%
Probit- generated, logit-fitted case	Sample size	700.00	21.7%	99.0%	13.87	[9.97, 17.68]	0.7%
Probit- generated, logit-fitted case	Sample size	1200.00	27.0%	100.0%	23.02	[18.52, 27.08]	0.0%
Forced-choice chance-floor case	Scale loca- tion	- 0.80	61.2%	76.4%	2.25	[0.19, 4.28]	37.4%
Forced-choice chance-floor case	Scale loca- tion	- 1.50	44.0%	70.4%	1.47	[-0.27, 3.47]	47.8%
Forced-choice chance-floor case	Scale loca- tion	0.00	52.0%	78.0%	2.39	[0.29, 4.56]	39.2%

Table 10: AIC comparison across the diagnostic atlas. Positive delta AIC favors the generating link. The within-two column reports the proportion of finite comparisons with absolute delta AIC below 2.

Case	Slice	Value	Pseudo-interaction detection	AIC favors generating link	Median delta AIC	Delta AIC IQR	Absolute delta AIC < 2
Forced-choice chance-floor case	Scale location	0.80	33.6%	73.4%	1.41	[-0.12, 2.96]	51.4%
Forced-choice chance-floor case	Scale location	1.50	9.0%	67.6%	0.63	[-0.30, 1.64]	76.2%
Probit-generated, logit-fitted case	Scale location	1.50	12.3%	93.0%	6.13	[3.81, 8.99]	13.3%
Probit-generated, logit-fitted case	Scale location	0.00	18.0%	95.0%	7.09	[4.06, 9.67]	10.0%
Probit-generated, logit-fitted case	Scale location	0.75	4.3%	99.0%	5.46	[3.54, 7.66]	12.0%
Probit-generated, logit-fitted case	Scale location	2.25	23.3%	93.0%	6.77	[3.73, 9.98]	10.7%

The AIC win rate is reported together with the magnitude of the difference because these describe different situations. A generating link that wins frequently by a very small margin is weaker evidence than one that wins by a large margin. As in the manuscript, these comparisons rank a limited candidate set. They do not identify the substantively correct scale by themselves.

7 What the atlas adds, and what it does not

The atlas strengthens the simulation evidence in three ways. First, it shows whether the manuscript anchors sit inside broader patterns or depend narrowly on one tuned configuration. Second, it separates parameters that change the **deterministic scale discrepancy**, such as scale location and main-effect magnitude, from parameters that mainly change **detectability**,

such as sample size and trial count. Third, the diagnostic extension places pseudo-interaction detection and model-checking performance on the same sensitivity slices.

The atlas does not estimate how often link misspecification occurs in psychological research. It does not cover every plausible outcome family, link, random-effects structure, measurement model, or predictor distribution. It also does not turn model diagnostics into a decision rule for the scientifically correct scale. Its role is narrower: to show how the mechanisms developed in the paper change across a declared set of nearby scenarios.

The Shiny companion can be used for optional exploration of these results. This document is the stable reporting layer: every figure and table is generated from the compact atlas summaries stored in `simulation-atlas/data/`.

8 Numerical appendix

The tables below provide a compact scenario-level record of the core atlas. The machine-readable CSV files remain the authoritative full output and contain additional design, Monte Carlo, convergence, and response-scale columns.

8.1 Forced-choice accuracy

Table 11: Compact scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
FC-001	paper anchor	Lower performance	chance-corrected logit	Chance-corrected binomial logit	3.6%	0.000	100.0%
FC-001	paper anchor	Lower performance	chance-corrected logit	Gaussian identity	41.8%	-0.077	100.0%
FC-001	paper anchor	Lower performance	chance-corrected logit	Standard binomial logit	55.4%	-0.453	100.0%
FC-001	paper anchor	Lower performance	chance-corrected logit	Standard binomial probit	53.4%	-0.262	100.0%

Table 11: Compact scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
FC-002	paper anchor	Middle performance	chance-corrected logit	Chance-corrected binomial logit	5.6%	0.000	100.0%
FC-002	paper anchor	Middle performance	chance-corrected logit	Gaussian identity	16.6%	-0.036	100.0%
FC-002	paper anchor	Middle performance	chance-corrected logit	Standard binomial logit	57.0%	-0.473	100.0%
FC-002	paper anchor	Middle performance	chance-corrected logit	Standard binomial probit	49.8%	-0.243	100.0%
FC-003	paper anchor	Higher performance	chance-corrected logit	Chance-corrected binomial logit	4.6%	0.000	100.0%
FC-003	paper anchor	Higher performance	chance-corrected logit	Gaussian identity	10.2%	0.028	100.0%
FC-003	paper anchor	Higher performance	chance-corrected logit	Standard binomial logit	36.8%	-0.396	100.0%
FC-003	paper anchor	Higher performance	chance-corrected logit	Standard binomial probit	21.2%	-0.153	100.0%
FC-004	main effect surface	Age effect x 0.50; group effect x 0.50	chance-corrected logit	Chance-corrected binomial logit	4.4%	0.000	100.0%
FC-004	main effect surface	Age effect x 0.50; group effect x 0.50	chance-corrected logit	Gaussian identity	5.0%	-0.007	100.0%
FC-004	main effect surface	Age effect x 0.50; group effect x 0.50	chance-corrected logit	Standard binomial logit	9.2%	-0.124	100.0%

Table 11: Compact scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
FC-004	main effect surface	Age effect x 0.50; group effect x 0.50	chance-corrected logit	Standard binomial probit	8.6%	-0.063	100.0%
FC-005	main effect surface	Age effect x 1.00; group effect x 0.50	chance-corrected logit	Chance-corrected binomial logit	4.4%	0.000	100.0%
FC-005	main effect surface	Age effect x 1.00; group effect x 0.50	chance-corrected logit	Gaussian identity	4.2%	-0.009	100.0%
FC-005	main effect surface	Age effect x 1.00; group effect x 0.50	chance-corrected logit	Standard binomial logit	15.0%	-0.232	100.0%
FC-005	main effect surface	Age effect x 1.00; group effect x 0.50	chance-corrected logit	Standard binomial probit	12.2%	-0.113	100.0%
FC-006	main effect surface	Age effect x 1.50; group effect x 0.50	chance-corrected logit	Chance-corrected binomial logit	5.6%	0.000	100.0%
FC-006	main effect surface	Age effect x 1.50; group effect x 0.50	chance-corrected logit	Gaussian identity	6.0%	-0.009	100.0%
FC-006	main effect surface	Age effect x 1.50; group effect x 0.50	chance-corrected logit	Standard binomial logit	25.4%	-0.313	100.0%
FC-006	main effect surface	Age effect x 1.50; group effect x 0.50	chance-corrected logit	Standard binomial probit	19.2%	-0.145	100.0%
FC-007	main effect surface	Age effect x 0.50; group effect x 1.00	chance-corrected logit	Chance-corrected binomial logit	5.0%	0.000	100.0%
FC-007	main effect surface	Age effect x 0.50; group effect x 1.00	chance-corrected logit	Gaussian identity	9.2%	-0.024	100.0%

Table 11: Compact scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
FC-007	main effect surface	Age effect x 0.50; group effect x 1.00	chance-corrected logit	Standard binomial logit	21.2%	-0.255	100.0%
FC-007	main effect surface	Age effect x 0.50; group effect x 1.00	chance-corrected logit	Standard binomial probit	18.0%	-0.136	100.0%
FC-008	main effect surface	Age effect x 1.50; group effect x 1.00	chance-corrected logit	Chance-corrected binomial logit	6.2%	0.000	100.0%
FC-008	main effect surface	Age effect x 1.50; group effect x 1.00	chance-corrected logit	Gaussian identity	21.8%	-0.034	100.0%
FC-008	main effect surface	Age effect x 1.50; group effect x 1.00	chance-corrected logit	Standard binomial logit	80.0%	-0.635	100.0%
FC-008	main effect surface	Age effect x 1.50; group effect x 1.00	chance-corrected logit	Standard binomial probit	71.8%	-0.309	100.0%
FC-009	main effect surface	Age effect x 0.50; group effect x 1.50	chance-corrected logit	Chance-corrected binomial logit	4.2%	0.000	100.0%
FC-009	main effect surface	Age effect x 0.50; group effect x 1.50	chance-corrected logit	Gaussian identity	17.8%	-0.048	100.0%
FC-009	main effect surface	Age effect x 0.50; group effect x 1.50	chance-corrected logit	Standard binomial logit	38.6%	-0.380	100.0%
FC-009	main effect surface	Age effect x 0.50; group effect x 1.50	chance-corrected logit	Standard binomial probit	33.2%	-0.210	100.0%
FC-010	main effect surface	Age effect x 1.00; group effect x 1.50	chance-corrected logit	Chance-corrected binomial logit	4.8%	0.000	100.0%

Table 11: Compact scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
FC-010	main effect surface	Age effect x 1.00; group effect x 1.50	chance-corrected logit	Gaussian identity	47.6%	-0.073	100.0%
FC-010	main effect surface	Age effect x 1.00; group effect x 1.50	chance-corrected logit	Standard binomial logit	89.0%	-0.706	100.0%
FC-010	main effect surface	Age effect x 1.00; group effect x 1.50	chance-corrected logit	Standard binomial probit	83.4%	-0.376	100.0%
FC-011	main effect surface	Age effect x 1.50; group effect x 1.50	chance-corrected logit	Chance-corrected binomial logit	2.8%	0.000	100.0%
FC-011	main effect surface	Age effect x 1.50; group effect x 1.50	chance-corrected logit	Gaussian identity	60.4%	-0.073	100.0%
FC-011	main effect surface	Age effect x 1.50; group effect x 1.50	chance-corrected logit	Standard binomial logit	98.4%	-0.949	100.0%
FC-011	main effect surface	Age effect x 1.50; group effect x 1.50	chance-corrected logit	Standard binomial probit	97.8%	-0.481	100.0%
FC-012	sample size	N = 50	chance-corrected logit	Chance-corrected binomial logit	3.6%	0.000	100.0%
FC-012	sample size	N = 50	chance-corrected logit	Gaussian identity	5.6%	-0.036	100.0%
FC-012	sample size	N = 50	chance-corrected logit	Standard binomial logit	16.0%	-0.473	100.0%
FC-012	sample size	N = 50	chance-corrected logit	Standard binomial probit	13.8%	-0.243	100.0%

Table 11: Compact scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo- interaction	Fit suc- cess
FC-013	sample size	N = 100	chance-corrected logit	Chance-corrected binomial logit	4.4%	0.000	100.0%
FC-013	sample size	N = 100	chance-corrected logit	Gaussian identity	7.8%	-0.036	100.0%
FC-013	sample size	N = 100	chance-corrected logit	Standard binomial logit	27.8%	-0.473	100.0%
FC-013	sample size	N = 100	chance-corrected logit	Standard binomial probit	23.0%	-0.243	100.0%
FC-014	sample size	N = 150	chance-corrected logit	Chance-corrected binomial logit	5.2%	0.000	100.0%
FC-014	sample size	N = 150	chance-corrected logit	Gaussian identity	10.8%	-0.036	100.0%
FC-014	sample size	N = 150	chance-corrected logit	Standard binomial logit	35.8%	-0.473	100.0%
FC-014	sample size	N = 150	chance-corrected logit	Standard binomial probit	30.2%	-0.243	100.0%
FC-015	sample size	N = 200	chance-corrected logit	Chance-corrected binomial logit	4.8%	0.000	100.0%
FC-015	sample size	N = 200	chance-corrected logit	Gaussian identity	10.6%	-0.036	100.0%
FC-015	sample size	N = 200	chance-corrected logit	Standard binomial logit	43.6%	-0.473	100.0%

Table 11: Compact scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo- interaction	Fit suc- cess
FC-015	sample size	$N = 200$	chance-corrected logit	Standard binomial probit	37.0%	-0.243	100.0%
FC-016	sample size	$N = 500$	chance-corrected logit	Chance-corrected binomial logit	4.4%	0.000	100.0%
FC-016	sample size	$N = 500$	chance-corrected logit	Gaussian identity	27.4%	-0.036	100.0%
FC-016	sample size	$N = 500$	chance-corrected logit	Standard binomial logit	85.2%	-0.473	100.0%
FC-016	sample size	$N = 500$	chance-corrected logit	Standard binomial probit	77.8%	-0.243	100.0%
FC-017	sample size	$N = 1000$	chance-corrected logit	Chance-corrected binomial logit	4.0%	0.000	100.0%
FC-017	sample size	$N = 1000$	chance-corrected logit	Gaussian identity	42.4%	-0.036	100.0%
FC-017	sample size	$N = 1000$	chance-corrected logit	Standard binomial logit	98.4%	-0.473	100.0%
FC-017	sample size	$N = 1000$	chance-corrected logit	Standard binomial probit	96.2%	-0.243	100.0%
FC-018	scale location	Intercept = -1.50	chance-corrected logit	Chance-corrected binomial logit	5.0%	0.000	100.0%
FC-018	scale location	Intercept = -1.50	chance-corrected logit	Gaussian identity	38.8%	-0.079	100.0%

Table 11: Compact scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo- interaction	Fit suc- cess
FC-018	scale location	Intercept = -1.50	chance- corrected logit	Standard binomial logit	44.0%	-0.365	100.0%
FC-018	scale location	Intercept = -1.50	chance- corrected logit	Standard binomial probit	43.8%	-0.222	100.0%
FC-019	scale location	Intercept = 1.50	chance- corrected logit	Chance- corrected binomial logit	4.8%	0.000	100.0%
FC-019	scale location	Intercept = 1.50	chance- corrected logit	Gaussian identity	66.2%	0.071	100.0%
FC-019	scale location	Intercept = 1.50	chance- corrected logit	Standard binomial logit	15.0%	-0.286	100.0%
FC-019	scale location	Intercept = 1.50	chance- corrected logit	Standard binomial probit	7.4%	-0.053	100.0%
FC-020	trial count	Trials = 5	chance- corrected logit	Chance- corrected binomial logit	5.2%	0.000	100.0%
FC-020	trial count	Trials = 5	chance- corrected logit	Gaussian identity	8.2%	-0.036	100.0%
FC-020	trial count	Trials = 5	chance- corrected logit	Standard binomial logit	22.0%	-0.473	100.0%
FC-020	trial count	Trials = 5	chance- corrected logit	Standard binomial probit	19.2%	-0.243	100.0%
FC-021	trial count	Trials = 10	chance- corrected logit	Chance- corrected binomial logit	5.0%	0.000	100.0%

Table 11: Compact scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
FC-021	trial count	Trials = 10	chance-corrected logit	Gaussian identity	12.2%	-0.036	100.0%
FC-021	trial count	Trials = 10	chance-corrected logit	Standard binomial logit	30.0%	-0.473	100.0%
FC-021	trial count	Trials = 10	chance-corrected logit	Standard binomial probit	25.6%	-0.243	100.0%
FC-022	trial count	Trials = 50	chance-corrected logit	Chance-corrected binomial logit	7.0%	0.000	100.0%
FC-022	trial count	Trials = 50	chance-corrected logit	Gaussian identity	34.2%	-0.036	100.0%
FC-022	trial count	Trials = 50	chance-corrected logit	Standard binomial logit	89.4%	-0.473	100.0%
FC-022	trial count	Trials = 50	chance-corrected logit	Standard binomial probit	84.0%	-0.243	100.0%
FC-023	chance level	Chance = 0.25	chance-corrected logit	Chance-corrected binomial logit	6.0%	0.000	100.0%
FC-023	chance level	Chance = 0.25	chance-corrected logit	Gaussian identity	25.4%	-0.054	100.0%
FC-023	chance level	Chance = 0.25	chance-corrected logit	Standard binomial logit	48.8%	-0.411	100.0%
FC-023	chance level	Chance = 0.25	chance-corrected logit	Standard binomial probit	45.6%	-0.228	100.0%

Table 11: Compact scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
FC-024	chance level	Chance = 0.3333	chance-corrected logit	Chance-corrected binomial logit	4.8%	0.000	100.0%
FC-024	chance level	Chance = 0.3333	chance-corrected logit	Gaussian identity	20.2%	-0.048	100.0%
FC-024	chance level	Chance = 0.3333	chance-corrected logit	Standard binomial logit	53.8%	-0.444	100.0%
FC-024	chance level	Chance = 0.3333	chance-corrected logit	Standard binomial probit	47.2%	-0.240	100.0%

8.2 Logit versus probit

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
WF-001	paper anchor	Logit-generated manuscript scenario	logit	Binomial logit	6.0%	0.000	100.0%
WF-001	paper anchor	Logit-generated manuscript scenario	logit	Binomial probit	17.3%	-0.112	100.0%
WF-002	paper anchor	Probit-generated manuscript scenario	probit	Binomial logit	14.3%	0.216	100.0%
WF-002	paper anchor	Probit-generated manuscript scenario	probit	Binomial probit	5.3%	0.000	100.0%
WF-003	main effect surface	logit generated: group x 0.50; condition x 0.50	logit	Binomial logit	6.0%	0.000	100.0%

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo- interaction	Fit suc- cess
WF-003	main effect surface	logit generated: group x 0.50; condition x 0.50	logit	Binomial probit	7.0%	-0.048	100.0%
WF-004	main effect surface	logit generated: group x 1.00; condition x 0.50	logit	Binomial logit	7.0%	0.000	100.0%
WF-004	main effect surface	logit generated: group x 1.00; condition x 0.50	logit	Binomial probit	10.0%	-0.080	100.0%
WF-005	main effect surface	logit generated: group x 1.50; condition x 0.50	logit	Binomial logit	4.3%	0.000	100.0%
WF-005	main effect surface	logit generated: group x 1.50; condition x 0.50	logit	Binomial probit	11.7%	-0.080	100.0%
WF-006	main effect surface	logit generated: group x 0.50; condition x 1.00	logit	Binomial logit	6.3%	0.000	100.0%
WF-006	main effect surface	logit generated: group x 0.50; condition x 1.00	logit	Binomial probit	12.0%	-0.080	100.0%
WF-007	main effect surface	logit generated: group x 1.50; condition x 1.00	logit	Binomial logit	5.7%	0.000	100.0%
WF-007	main effect surface	logit generated: group x 1.50; condition x 1.00	logit	Binomial probit	11.3%	-0.080	100.0%
WF-008	main effect surface	logit generated: group x 0.50; condition x 1.50	logit	Binomial logit	6.0%	0.000	100.0%
WF-008	main effect surface	logit generated: group x 0.50; condition x 1.50	logit	Binomial probit	10.7%	-0.080	100.0%
WF-009	main effect surface	logit generated: group x 1.00; condition x 1.50	logit	Binomial logit	6.0%	0.000	100.0%

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo- interaction	Fit suc- cess
WF-009	main effect surface	logit generated: group x 1.00; condition x 1.50	logit	Binomial probit	11.3%	-0.080	100.0%
WF-010	main effect surface	logit generated: group x 1.50; condition x 1.50	logit	Binomial logit	4.0%	0.000	100.0%
WF-010	main effect surface	logit generated: group x 1.50; condition x 1.50	logit	Binomial probit	5.3%	0.000	100.0%
WF-011	main effect surface	probit generated: group x 0.50; condition x 0.50	probit	Binomial logit	10.0%	0.107	100.0%
WF-011	main effect surface	probit generated: group x 0.50; condition x 0.50	probit	Binomial probit	6.7%	0.000	100.0%
WF-012	main effect surface	probit generated: group x 1.00; condition x 0.50	probit	Binomial logit	13.3%	0.162	100.0%
WF-012	main effect surface	probit generated: group x 1.00; condition x 0.50	probit	Binomial probit	5.3%	0.000	100.0%
WF-013	main effect surface	probit generated: group x 1.50; condition x 0.50	probit	Binomial logit	11.7%	0.162	100.0%
WF-013	main effect surface	probit generated: group x 1.50; condition x 0.50	probit	Binomial probit	4.3%	0.000	100.0%
WF-014	main effect surface	probit generated: group x 0.50; condition x 1.00	probit	Binomial logit	12.0%	0.162	100.0%
WF-014	main effect surface	probit generated: group x 0.50; condition x 1.00	probit	Binomial probit	4.0%	0.000	100.0%
WF-015	main effect surface	probit generated: group x 1.50; condition x 1.00	probit	Binomial logit	10.7%	0.162	100.0%

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo- interaction	Fit suc- cess
WF-015	main effect surface	probit generated: group x 1.50; condition x 1.00	probit	Binomial probit	5.3%	0.000	100.0%
WF-016	main effect surface	probit generated: group x 0.50; condition x 1.50	probit	Binomial logit	14.7%	0.162	100.0%
WF-016	main effect surface	probit generated: group x 0.50; condition x 1.50	probit	Binomial probit	6.3%	0.000	100.0%
WF-017	main effect surface	probit generated: group x 1.00; condition x 1.50	probit	Binomial logit	10.3%	0.162	100.0%
WF-017	main effect surface	probit generated: group x 1.00; condition x 1.50	probit	Binomial probit	6.3%	0.000	100.0%
WF-018	main effect surface	probit generated: group x 1.50; condition x 1.50	probit	Binomial logit	7.7%	0.000	100.0%
WF-018	main effect surface	probit generated: group x 1.50; condition x 1.50	probit	Binomial probit	7.7%	0.000	100.0%
WF-019	sample size	logit generated: subjects = 100	logit	Binomial logit	5.3%	0.000	100.0%
WF-019	sample size	logit generated: subjects = 100	logit	Binomial probit	7.3%	-0.112	100.0%
WF-020	sample size	logit generated: subjects = 200	logit	Binomial logit	5.3%	0.000	100.0%
WF-020	sample size	logit generated: subjects = 200	logit	Binomial probit	8.7%	-0.112	100.0%
WF-021	sample size	logit generated: subjects = 300	logit	Binomial logit	7.7%	0.000	100.0%
WF-021	sample size	logit generated: subjects = 300	logit	Binomial probit	13.0%	-0.112	100.0%
WF-022	sample size	logit generated: subjects = 400	logit	Binomial logit	2.7%	0.000	100.0%

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo- interaction	Fit suc- cess
WF-022	sample size	logit generated: subjects = 400	logit	Binomial probit	9.7%	-0.112	100.0%
WF-023	sample size	logit generated: subjects = 1200	logit	Binomial logit	5.3%	0.000	100.0%
WF-023	sample size	logit generated: subjects = 1200	logit	Binomial probit	20.3%	-0.112	100.0%
WF-024	sample size	logit generated: subjects = 2000	logit	Binomial logit	5.3%	0.000	100.0%
WF-024	sample size	logit generated: subjects = 2000	logit	Binomial probit	37.0%	-0.112	100.0%
WF-025	trial count	logit generated: trials per cell = 5	logit	Binomial logit	4.0%	0.000	100.0%
WF-025	trial count	logit generated: trials per cell = 5	logit	Binomial probit	7.3%	-0.112	100.0%
WF-026	trial count	logit generated: trials per cell = 30	logit	Binomial logit	6.0%	0.000	100.0%
WF-026	trial count	logit generated: trials per cell = 30	logit	Binomial probit	28.3%	-0.112	100.0%
WF-027	latent location	logit generated: reference intercept = 0	logit	Binomial logit	6.7%	0.000	100.0%
WF-027	latent location	logit generated: reference intercept = 0	logit	Binomial probit	40.3%	0.175	100.0%
WF-028	latent location	logit generated: reference intercept = 0.75	logit	Binomial logit	5.0%	0.000	100.0%
WF-028	latent location	logit generated: reference intercept = 0.75	logit	Binomial probit	9.0%	0.059	100.0%
WF-029	latent location	logit generated: reference intercept = 2.25	logit	Binomial logit	5.0%	0.000	100.0%

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo- interaction	Fit suc- cess
WF-029	latent location	logit generated: reference intercept = 2.25	logit	Binomial probit	46.0%	-0.186	100.0%
WF-030	icc	logit generated: target ICC = 0	logit	Binomial logit	4.7%	0.000	100.0%
WF-030	icc	logit generated: target ICC = 0	logit	Binomial probit	79.0%	-0.112	100.0%
WF-031	icc	logit generated: target ICC = 0.15	logit	Binomial logit	7.3%	0.000	100.0%
WF-031	icc	logit generated: target ICC = 0.15	logit	Binomial probit	37.0%	-0.112	100.0%
WF-032	icc	logit generated: target ICC = 0.45	logit	Binomial logit	7.3%	0.000	100.0%
WF-032	icc	logit generated: target ICC = 0.45	logit	Binomial probit	8.7%	-0.112	100.0%
WF-033	sample size	probit generated: subjects = 100	probit	Binomial logit	7.7%	0.216	100.0%
WF-033	sample size	probit generated: subjects = 100	probit	Binomial probit	5.0%	0.000	100.0%
WF-034	sample size	probit generated: subjects = 200	probit	Binomial logit	8.0%	0.216	100.0%
WF-034	sample size	probit generated: subjects = 200	probit	Binomial probit	4.7%	0.000	100.0%
WF-035	sample size	probit generated: subjects = 300	probit	Binomial logit	15.3%	0.216	100.0%
WF-035	sample size	probit generated: subjects = 300	probit	Binomial probit	5.0%	0.000	100.0%
WF-036	sample size	probit generated: subjects = 400	probit	Binomial logit	13.7%	0.216	100.0%
WF-036	sample size	probit generated: subjects = 400	probit	Binomial probit	5.7%	0.000	100.0%
WF-037	sample size	probit generated: subjects = 1200	probit	Binomial logit	28.7%	0.216	100.0%

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo- interaction	Fit suc- cess
WF-037	sample size	probit generated: subjects = 1200	probit	Binomial probit	4.7%	0.000	100.0%
WF-038	sample size	probit generated: subjects = 2000	probit	Binomial logit	43.0%	0.216	100.0%
WF-038	sample size	probit generated: subjects = 2000	probit	Binomial probit	4.0%	0.000	100.0%
WF-039	trial count	probit generated: trials per cell = 5	probit	Binomial logit	9.7%	0.216	100.0%
WF-039	trial count	probit generated: trials per cell = 5	probit	Binomial probit	6.0%	0.000	100.0%
WF-040	trial count	probit generated: trials per cell = 30	probit	Binomial logit	31.7%	0.216	100.0%
WF-040	trial count	probit generated: trials per cell = 30	probit	Binomial probit	6.0%	0.000	100.0%
WF-041	latent location	probit generated: reference intercept = 0	probit	Binomial logit	40.0%	-0.424	100.0%
WF-041	latent location	probit generated: reference intercept = 0	probit	Binomial probit	6.0%	0.000	100.0%
WF-042	latent location	probit generated: reference intercept = 0.75	probit	Binomial logit	7.7%	-0.108	100.0%
WF-042	latent location	probit generated: reference intercept = 0.75	probit	Binomial probit	3.7%	0.000	100.0%
WF-043	latent location	probit generated: reference intercept = 2.25	probit	Binomial logit	44.3%	0.520	100.0%
WF-043	latent location	probit generated: reference intercept = 2.25	probit	Binomial probit	5.7%	0.000	100.0%
WF-044	icc	probit generated: target ICC = 0	probit	Binomial logit	82.3%	0.216	100.0%

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
WF-044	icc	probit generated: target ICC = 0	probit	Binomial probit	6.0%	0.000	100.0%
WF-045	icc	probit generated: target ICC = 0.15	probit	Binomial logit	38.3%	0.216	100.0%
WF-045	icc	probit generated: target ICC = 0.15	probit	Binomial probit	3.0%	0.000	100.0%
WF-046	icc	probit generated: target ICC = 0.45	probit	Binomial logit	8.3%	0.216	100.0%
WF-046	icc	probit generated: target ICC = 0.45	probit	Binomial probit	6.3%	0.000	100.0%

8.3 Sum scores

Table 13: Compact scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
SS-001	paper anchor	Lower range	graded logistic latent scale	Gaussian-probit bounded score	6.6%	-0.056	100.0%
SS-001	paper anchor	Lower range	graded logistic latent scale	Latent generating scale	3.6%	0.000	100.0%
SS-001	paper anchor	Lower range	graded logistic latent scale	Observed sum-score identity	72.6%	-2.034	100.0%
SS-002	paper anchor	Middle range	graded logistic latent scale	Gaussian-probit bounded score	3.8%	-0.010	100.0%

Table 13: Compact scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
SS-002	paper anchor	Middle range	graded logistic latent scale	Latent generating scale	5.8%	0.000	100.0%
SS-002	paper anchor	Middle range	graded logistic latent scale	Observed sum-score identity	7.4%	-0.696	100.0%
SS-003	paper anchor	Upper range	graded logistic latent scale	Gaussian-probit bounded score	5.6%	0.027	100.0%
SS-003	paper anchor	Upper range	graded logistic latent scale	Latent generating scale	4.4%	0.000	100.0%
SS-003	paper anchor	Upper range	graded logistic latent scale	Observed sum-score identity	37.6%	1.570	100.0%
SS-004	main effect surface	x effect x 0.50; group effect x 0.50	graded logistic latent scale	Gaussian-probit bounded score	4.6%	-0.001	100.0%
SS-004	main effect surface	x effect x 0.50; group effect x 0.50	graded logistic latent scale	Latent generating scale	3.2%	0.000	100.0%
SS-004	main effect surface	x effect x 0.50; group effect x 0.50	graded logistic latent scale	Observed sum-score identity	5.2%	-0.096	100.0%
SS-005	main effect surface	x effect x 1.00; group effect x 0.50	graded logistic latent scale	Gaussian-probit bounded score	2.6%	-0.002	100.0%
SS-005	main effect surface	x effect x 1.00; group effect x 0.50	graded logistic latent scale	Latent generating scale	3.6%	0.000	100.0%
SS-005	main effect surface	x effect x 1.00; group effect x 0.50	graded logistic latent scale	Observed sum-score identity	2.8%	-0.180	100.0%

Table 13: Compact scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
SS-006	main effect surface	x effect x 1.50; group effect x 0.50	graded logistic latent scale	Gaussian-probit bounded score	3.8%	-0.004	100.0%
SS-006	main effect surface	x effect x 1.50; group effect x 0.50	graded logistic latent scale	Latent generating scale	6.0%	0.000	100.0%
SS-006	main effect surface	x effect x 1.50; group effect x 0.50	graded logistic latent scale	Observed sum-score identity	3.4%	-0.242	100.0%
SS-007	main effect surface	x effect x 0.50; group effect x 1.00	graded logistic latent scale	Gaussian-probit bounded score	5.6%	-0.005	100.0%
SS-007	main effect surface	x effect x 0.50; group effect x 1.00	graded logistic latent scale	Latent generating scale	5.8%	0.000	100.0%
SS-007	main effect surface	x effect x 0.50; group effect x 1.00	graded logistic latent scale	Observed sum-score identity	6.0%	-0.370	100.0%
SS-008	main effect surface	x effect x 1.50; group effect x 1.00	graded logistic latent scale	Gaussian-probit bounded score	3.8%	-0.015	100.0%
SS-008	main effect surface	x effect x 1.50; group effect x 1.00	graded logistic latent scale	Latent generating scale	6.8%	0.000	100.0%
SS-008	main effect surface	x effect x 1.50; group effect x 1.00	graded logistic latent scale	Observed sum-score identity	11.0%	-0.940	100.0%
SS-009	main effect surface	x effect x 0.50; group effect x 1.50	graded logistic latent scale	Gaussian-probit bounded score	3.4%	-0.011	100.0%

Table 13: Compact scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
SS-009	main effect surface	x effect x 0.50; group effect x 1.50	graded logistic latent scale	Latent generating scale	4.0%	0.000	100.0%
SS-009	main effect surface	x effect x 0.50; group effect x 1.50	graded logistic latent scale	Observed sum-score identity	11.6%	-0.786	100.0%
SS-010	main effect surface	x effect x 1.00; group effect x 1.50	graded logistic latent scale	Gaussian-probit bounded score	4.8%	-0.022	100.0%
SS-010	main effect surface	x effect x 1.00; group effect x 1.50	graded logistic latent scale	Latent generating scale	5.4%	0.000	100.0%
SS-010	main effect surface	x effect x 1.00; group effect x 1.50	graded logistic latent scale	Observed sum-score identity	29.8%	-1.481	100.0%
SS-011	main effect surface	x effect x 1.50; group effect x 1.50	graded logistic latent scale	Gaussian-probit bounded score	3.6%	-0.036	100.0%
SS-011	main effect surface	x effect x 1.50; group effect x 1.50	graded logistic latent scale	Latent generating scale	4.0%	0.000	100.0%
SS-011	main effect surface	x effect x 1.50; group effect x 1.50	graded logistic latent scale	Observed sum-score identity	41.0%	-2.008	100.0%
SS-012	sample size	N = 75	graded logistic latent scale	Gaussian-probit bounded score	3.2%	-0.010	100.0%
SS-012	sample size	N = 75	graded logistic latent scale	Latent generating scale	5.2%	0.000	100.0%
SS-012	sample size	N = 75	graded logistic latent scale	Observed sum-score identity	4.2%	-0.696	100.0%

Table 13: Compact scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo- interaction	Fit suc- cess
SS-013	sample size	N = 150	graded logistic latent scale	Gaussian-probit bounded score	4.4%	-0.010	100.0%
SS-013	sample size	N = 150	graded logistic latent scale	Latent generating scale	7.0%	0.000	100.0%
SS-013	sample size	N = 150	graded logistic latent scale	Observed sum-score identity	3.6%	-0.696	100.0%
SS-014	sample size	N = 225	graded logistic latent scale	Gaussian-probit bounded score	3.8%	-0.010	100.0%
SS-014	sample size	N = 225	graded logistic latent scale	Latent generating scale	5.0%	0.000	100.0%
SS-014	sample size	N = 225	graded logistic latent scale	Observed sum-score identity	5.0%	-0.696	100.0%
SS-015	sample size	N = 300	graded logistic latent scale	Gaussian-probit bounded score	5.2%	-0.010	100.0%
SS-015	sample size	N = 300	graded logistic latent scale	Latent generating scale	4.4%	0.000	100.0%
SS-015	sample size	N = 300	graded logistic latent scale	Observed sum-score identity	6.6%	-0.696	100.0%
SS-016	sample size	N = 450	graded logistic latent scale	Gaussian-probit bounded score	4.4%	-0.010	100.0%

Table 13: Compact scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
SS-016	sample size	N = 450	graded logistic latent scale	Latent generating scale	5.2%	0.000	100.0%
SS-016	sample size	N = 450	graded logistic latent scale	Observed sum-score identity	8.4%	-0.696	100.0%
SS-017	sample size	N = 1200	graded logistic latent scale	Gaussian-probit bounded score	2.6%	-0.010	100.0%
SS-017	sample size	N = 1200	graded logistic latent scale	Latent generating scale	6.0%	0.000	100.0%
SS-017	sample size	N = 1200	graded logistic latent scale	Observed sum-score identity	11.0%	-0.696	100.0%
SS-018	scale location	Threshold shift = -2.00	graded logistic latent scale	Gaussian-probit bounded score	5.8%	0.038	100.0%
SS-018	scale location	Threshold shift = -2.00	graded logistic latent scale	Latent generating scale	6.4%	0.000	100.0%
SS-018	scale location	Threshold shift = -2.00	graded logistic latent scale	Observed sum-score identity	55.8%	1.877	100.0%
SS-019	scale location	Threshold shift = 2.00	graded logistic latent scale	Gaussian-probit bounded score	12.8%	-0.072	100.0%
SS-019	scale location	Threshold shift = 2.00	graded logistic latent scale	Latent generating scale	3.0%	0.000	100.0%
SS-019	scale location	Threshold shift = 2.00	graded logistic latent scale	Observed sum-score identity	83.2%	-1.993	100.0%

Table 13: Compact scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit success
SS-020	item count	Items = 5	graded logistic latent scale	Gaussian-probit bounded score	2.6%	-0.013	100.0%
SS-020	item count	Items = 5	graded logistic latent scale	Latent generating scale	3.0%	0.000	100.0%
SS-020	item count	Items = 5	graded logistic latent scale	Observed sum-score identity	8.2%	-0.384	100.0%
SS-021	item count	Items = 15	graded logistic latent scale	Gaussian-probit bounded score	3.2%	-0.008	100.0%
SS-021	item count	Items = 15	graded logistic latent scale	Latent generating scale	5.4%	0.000	100.0%
SS-021	item count	Items = 15	graded logistic latent scale	Observed sum-score identity	11.2%	-1.162	100.0%
SS-022	latent dispersion	Latent SD = 0.70	graded logistic latent scale	Gaussian-probit bounded score	4.2%	-0.013	100.0%
SS-022	latent dispersion	Latent SD = 0.70	graded logistic latent scale	Latent generating scale	3.8%	0.000	100.0%
SS-022	latent dispersion	Latent SD = 0.70	graded logistic latent scale	Observed sum-score identity	11.8%	-0.831	100.0%
SS-023	latent dispersion	Latent SD = 1.30	graded logistic latent scale	Gaussian-probit bounded score	3.8%	-0.007	100.0%

Table 13: Compact scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo- interaction	Fit suc- cess
SS-023	latent dispersion	Latent SD = 1.30	graded logistic latent scale	Latent generating scale	4.6%	0.000	100.0%
SS-023	latent dispersion	Latent SD = 1.30	graded logistic latent scale	Observed sum-score identity	6.4%	-0.563	100.0%